C

CAMBRIDGE ADDITIONAL MATHEMATICS (0606) • UNIT 4.01

Logarithmic & Exponential Functions

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	**Q2** A	**Q3** A	**Q4** A	**Q5** B
Q6 A	**Q7** A	**Q8** A	**Q9** A	**Q10** A
Q11 D	**Q12** D	**Q13** B	**Q14** A	**Q15** B
Q16 D	**Q17** B	**Q18** C	**Q19** C	**Q20** D
Q21 A	**Q22** B	**Q23** B	**Q24** C	**Q25** B

Logarithmic & Exponential Functions Practice Exercises

1 MCQ

Solve the exponential equation $2^{x+2} = 5^x$, giving x in terms of common logarithms.

- **A** $x = \frac{2\ln 2}{\ln 5 - \ln 2}$ ✓
- **B** $x = \frac{2\ln 5}{\ln 5 - \ln 2}$
- **C** $x = \frac{\ln 2}{2(\ln 5 - \ln 2)}$
- **D** $x = \frac{2}{\ln 5 - \ln 2}$

WORKED SOLUTION & VERIFICATION

Take natural logs of both sides: $(x+2)\ln 2 = x\ln 5 \implies x\ln 2 + 2\ln 2 = x\ln 5 \implies x(\ln 5 - \ln 2) = 2\ln 2 \implies x = \frac{2\ln 2}{\ln 5 - \ln 2}$. [M1][A1]

2 MCQ

Solve the exponential equation $2^{x+3} = 5^x$, giving x in terms of common logarithms.

- **A** $x = \frac{3\ln 2}{\ln 5 - \ln 2}$ ✓
- **B** $x = \frac{3\ln 5}{\ln 5 - \ln 2}$
- **C** $x = \frac{3}{\ln 5 - \ln 2}$
- **D** $x = \frac{\ln 2}{3(\ln 5 - \ln 2)}$

WORKED SOLUTION & VERIFICATION

Take natural logs of both sides: $(x+3)\ln 2 = x\ln 5 \implies x\ln 2 + 3\ln 2 = x\ln 5 \implies x(\ln 5 - \ln 2) = 3\ln 2 \implies x = \frac{3\ln 2}{\ln 5 - \ln 2}$. [M1][A1]

3 MCQ

Solve the exponential equation $2^{x+4} = 5^x$, giving x in terms of common logarithms.

- **A** $x = \frac{4\ln 2}{\ln 5 - \ln 2}$ ✓
- **B** $x = \frac{\ln 2}{4(\ln 5 - \ln 2)}$
- **C** $x = \frac{4\ln 5}{\ln 5 - \ln 2}$
- **D** $x = \frac{4}{\ln 5 - \ln 2}$

WORKED SOLUTION & VERIFICATION

Take natural logs of both sides: $(x+4)\ln 2 = x\ln 5 \implies x\ln 2 + 4\ln 2 = x\ln 5 \implies x(\ln 5 - \ln 2) = 4\ln 2 \implies x = \frac{4\ln 2}{\ln 5 - \ln 2}$. [M1][A1]

4 MCQ

Solve the exponential equation $2^{x+5} = 5^x$, giving x in terms of common logarithms.

- **A** $x = \frac{5\ln 2}{\ln 5 - \ln 2}$ ✓
- **B** $x = \frac{5\ln 5}{\ln 5 - \ln 2}$
- **C** $x = \frac{5}{\ln 5 - \ln 2}$
- **D** $x = \frac{\ln 2}{5(\ln 5 - \ln 2)}$

WORKED SOLUTION & VERIFICATION

Take natural logs of both sides: $(x+5)\ln 2 = x\ln 5 \implies x\ln 2 + 5\ln 2 = x\ln 5 \implies x(\ln 5 - \ln 2) = 5\ln 2 \implies x = \frac{5\ln 2}{\ln 5 - \ln 2}$. [M1][A1]

5 MCQ

Solve the exponential equation $2^{x+6} = 5^x$, giving x in terms of common logarithms.

- **A** $x = \frac{6\ln 5}{\ln 5 - \ln 2}$
- **B** $x = \frac{6\ln 2}{\ln 5 - \ln 2}$ ✓
- **C** $x = \frac{6}{\ln 5 - \ln 2}$
- **D** $x = \frac{\ln 2}{6(\ln 5 - \ln 2)}$

WORKED SOLUTION & VERIFICATION

Take natural logs of both sides: $(x+6)\ln 2 = x\ln 5 \implies x\ln 2 + 6\ln 2 = x\ln 5 \implies x(\ln 5 - \ln 2) = 6\ln 2 \implies x = \frac{6\ln 2}{\ln 5 - \ln 2}$. [M1][A1]

6 MCQ

Solve the exponential equation $2^{x+7} = 5^x$, giving x in terms of common logarithms.

- **A** $x = \frac{7\ln 2}{\ln 5 - \ln 2}$ ✓
- **B** $x = \frac{\ln 2}{7(\ln 5 - \ln 2)}$
- **C** $x = \frac{7\ln 5}{\ln 5 - \ln 2}$
- **D** $x = \frac{7}{\ln 5 - \ln 2}$

WORKED SOLUTION & VERIFICATION

Take natural logs of both sides: $(x+7)\ln 2 = x\ln 5 \implies x\ln 2 + 7\ln 2 = x\ln 5 \implies x(\ln 5 - \ln 2) = 7\ln 2 \implies x = \frac{7\ln 2}{\ln 5 - \ln 2}$. [M1][A1]

7 MCQ

Solve the exponential equation $2^{x+8} = 5^x$, giving x in terms of common logarithms.

- **A** $x = \frac{8\ln 2}{\ln 5 - \ln 2}$ ✓
- **B** $x = \frac{8\ln 5}{\ln 5 - \ln 2}$
- **C** $x = \frac{\ln 2}{8(\ln 5 - \ln 2)}$
- **D** $x = \frac{8}{\ln 5 - \ln 2}$

WORKED SOLUTION & VERIFICATION

Take natural logs of both sides: $(x+8)\ln 2 = x\ln 5 \implies x\ln 2 + 8\ln 2 = x\ln 5 \implies x(\ln 5 - \ln 2) = 8\ln 2 \implies x = \frac{8\ln 2}{\ln 5 - \ln 2}$. [M1][A1]

8 MCQ

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 3$. What is the value of x?

- **A** 12 ✓
- **B** 3
- **C** 6
- **D** 14

WORKED SOLUTION & VERIFICATION

From the first equation: $\log_2(x/y) = 2 \implies x/y = 2^2 = 4 \implies x = 4y$. Substitute into second: $4y - 3y = 3 \implies y = 3$. Thus $x = 4(3) = 12$. [M1][A1]

9 MCQ

Find the root(s) satisfying the exponential equation $2^{x+3} = 5^x$, giving x in terms of common logarithms.

- **A** $x = \frac{3\ln 2}{\ln 5 - \ln 2}$ ✓
- **B** $x = \frac{3\ln 5}{\ln 5 - \ln 2}$
- **C** $x = \frac{3}{\ln 5 - \ln 2}$
- **D** $x = \frac{\ln 2}{3(\ln 5 - \ln 2)}$

WORKED SOLUTION & VERIFICATION

Take natural logs of both sides: $(x+3)\ln 2 = x\ln 5 \implies x\ln 2 + 3\ln 2 = x\ln 5 \implies x(\ln 5 - \ln 2) = 3\ln 2 \implies x = \frac{3\ln 2}{\ln 5 - \ln 2}$. [M1][A1]

10 MCQ

Calculate the solution set for the exponential equation $2^{x+4} = 5^x$, giving x in terms of common logarithms.

- **A** $x = \frac{4\ln 2}{\ln 5 - \ln 2}$ ✓
- **B** $x = \frac{\ln 2}{4(\ln 5 - \ln 2)}$
- **C** $x = \frac{4\ln 5}{\ln 5 - \ln 2}$
- **D** $x = \frac{4}{\ln 5 - \ln 2}$

WORKED SOLUTION & VERIFICATION

Take natural logs of both sides: $(x+4)\ln 2 = x\ln 5 \implies x\ln 2 + 4\ln 2 = x\ln 5 \implies x(\ln 5 - \ln 2) = 4\ln 2 \implies x = \frac{4\ln 2}{\ln 5 - \ln 2}$. [M1][A1]

11 MCQ

Obtain the value(s) satisfying the exponential equation $2^{x+5} = 5^x$, giving x in terms of common logarithms.

- **A** $x = \frac{5\ln 5}{\ln 5 - \ln 2}$
- **B** $x = \frac{5}{\ln 5 - \ln 2}$
- **C** $x = \frac{\ln 2}{5(\ln 5 - \ln 2)}$
- **D** $x = \frac{5\ln 2}{\ln 5 - \ln 2}$ ✓

WORKED SOLUTION & VERIFICATION

Take natural logs of both sides: $(x+5)\ln 2 = x\ln 5 \implies x\ln 2 + 5\ln 2 = x\ln 5 \implies x(\ln 5 - \ln 2) = 5\ln 2 \implies x = \frac{5\ln 2}{\ln 5 - \ln 2}$. [M1][A1]

12 MCQ

Work out the solution to the exponential equation $2^{x+6} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{6}{\ln 5 - \ln 2}$

B $x = \frac{\ln 2}{6(\ln 5 - \ln 2)}$

C $x = \frac{6 \ln 5}{\ln 5 - \ln 2}$

D $x = \frac{6 \ln 2}{\ln 5 - \ln 2}$ ✓

WORKED SOLUTION & VERIFICATION

Take natural logs of both sides: $(x+6)\ln 2 = x \ln 5 \implies x\ln 2 + 6\ln 2 = x\ln 5 \implies x(\ln 5 - \ln 2) = 6\ln 2 \implies x = \frac{6\ln 2}{\ln 5 - \ln 2}$. [M1][A1]

13 MCQ

Determine the solution to the exponential equation $2^{x+7} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{7 \ln 5}{\ln 5 - \ln 2}$

B $x = \frac{7 \ln 2}{\ln 5 - \ln 2}$ ✓

C $x = \frac{7}{\ln 5 - \ln 2}$

D $x = \frac{\ln 2}{7(\ln 5 - \ln 2)}$

WORKED SOLUTION & VERIFICATION

Take natural logs of both sides: $(x+7)\ln 2 = x \ln 5 \implies x\ln 2 + 7\ln 2 = x\ln 5 \implies x(\ln 5 - \ln 2) = 7\ln 2 \implies x = \frac{7\ln 2}{\ln 5 - \ln 2}$. [M1][A1]

14 MCQ

Find the root(s) satisfying the exponential equation $2^{x+8} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{8 \ln 2}{\ln 5 - \ln 2}$ ✓

B $x = \frac{8 \ln 5}{\ln 5 - \ln 2}$

C $x = \frac{\ln 2}{8(\ln 5 - \ln 2)}$

D $x = \frac{8}{\ln 5 - \ln 2}$

WORKED SOLUTION & VERIFICATION

Take natural logs of both sides: $(x+8)\ln 2 = x \ln 5 \implies x\ln 2 + 8\ln 2 = x\ln 5 \implies x(\ln 5 - \ln 2) = 8\ln 2 \implies x = \frac{8\ln 2}{\ln 5 - \ln 2}$. [M1][A1]

15 MCQ

Find the root(s) satisfying the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 4$. What is the value of x?

A 8

B 16 ✓

C 4

D 18

WORKED SOLUTION & VERIFICATION

From the first equation: $\log_2(x/y) = 2 \implies x/y = 2^2 = 4 \implies x = 4y$. Substitute into second: $4y - 3y = 4 \implies y = 4$. Thus $x = 4(4) = 16$. [M1][A1]

16 MCQ

Calculate the solution set for the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 5$. What is the value of x?

A 5

B 22

C 10

D **20** ✓

WORKED SOLUTION & VERIFICATION

From the first equation: $\log_2(x/y) = 2 \implies x/y = 2^2 = 4 \implies x = 4y$. Substitute into second: $4y - 3y = 5 \implies y = 5$. Thus $x = 4(5) = 20$. [M1][A1]

17 MCQ

Obtain the value(s) satisfying the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 6$. What is the value of x?

A 12

B **24** ✓

C 6

D 26

WORKED SOLUTION & VERIFICATION

From the first equation: $\log_2(x/y) = 2 \implies x/y = 2^2 = 4 \implies x = 4y$. Substitute into second: $4y - 3y = 6 \implies y = 6$. Thus $x = 4(6) = 24$. [M1][A1]

18 MCQ

Work out the solution to the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 7$. What is the value of x?

A 14

B 30

C **28** ✓

D 7

WORKED SOLUTION & VERIFICATION

From the first equation: $\log_2(x/y) = 2 \implies x/y = 2^2 = 4 \implies x = 4y$. Substitute into second: $4y - 3y = 7 \implies y = 7$. Thus $x = 4(7) = 28$. [M1][A1]

19 MCQ

Determine the solution to the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 8$. What is the value of x?

A 16

B 8

C **32** ✓

D 34

WORKED SOLUTION & VERIFICATION

From the first equation: $\log_2(x/y) = 2 \implies x/y = 2^2 = 4 \implies x = 4y$. Substitute into second: $4y - 3y = 8 \implies y = 8$. Thus $x = 4(8) = 32$. [M1][A1]

20 MCQ

Find the root(s) satisfying the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 9$. What is the value of x?

A 9

B 38

C 18

D 36 ✓

WORKED SOLUTION & VERIFICATION

From the first equation: $\log_2(x/y) = 2 \implies x/y = 2^2 = 4 \implies x = 4y$. Substitute into second: $4y - 3y = 9 \implies y = 9$. Thus $x = 4(9) = 36$. [M1][A1]

21 MCQ

Variables x and y satisfy $y = Ae^{bx}$. When $\ln y$ is plotted against x (Experiment #1), a straight line with gradient 2 and vertical intercept 1 is obtained. Find the exact value of y when $x = 2$.

A e^5 ✓

B $1e^2$

C e^4

D 5

WORKED SOLUTION & VERIFICATION

Taking natural logs gives $\ln y = \ln A + bx$, so $\ln A = 1$ and $b = 2$. At $x = 2$, $\ln y = 1 + (2)(2) = 5$, hence $y = e^5$.

22 MCQ

Variables x and y satisfy $y = Ae^{bx}$. When $\ln y$ is plotted against x (Experiment #2), a straight line with gradient -3 and vertical intercept 1 is obtained. Find the exact value of y when $x = 3$.

A e^0

B e^{-8} ✓

C -8

D $1e^{-3}$

WORKED SOLUTION & VERIFICATION

Taking natural logs gives $\ln y = \ln A + bx$, so $\ln A = 1$ and $b = -3$. At $x = 3$, $\ln y = 1 + (-3)(3) = -8$, hence $y = e^{-8}$.

23 MCQ

Variables x and y satisfy $y = Ae^{bx}$. When $\ln y$ is plotted against x (Experiment #3), a straight line with gradient 4 and vertical intercept 1 is obtained. Find the exact value of y when $x = 4$.

A 17

B e^{17} ✓

C $1e^4$

D e^8

WORKED SOLUTION & VERIFICATION

Taking natural logs gives $\ln y = \ln A + bx$, so $\ln A = 1$ and $b = 4$. At $x = 4$, $\ln y = 1 + (4)(4) = 17$, hence $y = e^{17}$.

24 MCQ

Variables x and y satisfy $y = Ae^{bx}$. When $\ln y$ is plotted against x (Experiment #4), a straight line with gradient -5 and vertical intercept 1 is obtained. Find the exact value of y when $x = 5$.

A -24

B $1e^{-5}$

C e^{-24} ✓

D e^{0}

WORKED SOLUTION & VERIFICATION

Taking natural logs gives $\ln y = \ln A + bx$, so $\ln A = 1$ and $b = -5$. At $x = 5$, $\ln y = 1 + (-5)(5) = -24$, hence $y = e^{-24}$.

25 MCQ

Variables x and y satisfy $y = Ae^{bx}$. When $\ln y$ is plotted against x (Experiment #5), a straight line with gradient 6 and vertical intercept 1 is obtained. Find the exact value of y when $x = 2$.

A e^{8}

B e^{13} ✓

C $1e^{6}$

D 13

WORKED SOLUTION & VERIFICATION

Taking natural logs gives $\ln y = \ln A + bx$, so $\ln A = 1$ and $b = 6$. At $x = 2$, $\ln y = 1 + (6)(2) = 13$, hence $y = e^{13}$.